

How many dimensions does a twelve-rung HAR ladder carry?

Method note: one column matches twelve, and the rank-3 win was one window

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Abstract

A base-2 HAR ladder holds twelve moving averages of one series at horizons $\{1, 2, \dots, 2048\}$. We ask how many of those twelve are doing work. The answer is one. A single fixed column—no estimation, no fitted direction—matches all twelve rungs, by three independent routes that agree: a nine-window placebo averages $+0.00001$, the pooled estimated direction gives -0.00054 ($t = -1.33$), and the best *a priori* shape gives $+0.00034$ ($t = +1.65$). The mechanism is that the ladder’s effective rank is 2.81 of 12 with 72.8% of its variance in the top direction, so eleven of the twelve columns are near-duplicates. Which single column works is governed by one parameter. Because rung j is a block mean of the last 2^j squared returns, a weight $w_j \propto L_j^{-\alpha}$ over rungs induces a kernel $\kappa(i) \propto i^{-(\alpha+1)}$ over *raw* lags, so α is the long-memory parameter d ; kernels decaying slower than $i^{-1.35}$ fail in 0–2 eras of 8, and everything from $d \approx 0.4$ upward ties. That optimum agrees with the standard estimate for realised variance, which is independent corroboration rather than a fitted result. An earlier version of this note reported that *three* supervised coefficients beat twelve rungs by 0.00080. That claim is withdrawn on evidence far stronger than the original doubt: repeating the identical procedure on the eight estimation windows nobody chose puts all eight behind the full ladder in a tight distribution (mean $+0.00022$, sd 0.00016), with the published window 13.5 standard deviations outside it. It is not a chance draw, not curvature, and not a discoverable feature—every imposed shape fails, including one hand-copied from the winning window’s own loadings. §5 lists five statements this note previously made that were wrong.

1 Panel and rows

Cache `results/b2_mmap_warm`, the corrected-fill build: availability recorded *before* the forward fill, a 26-bar staleness bound, scale estimated from observed rows only, and a warm-up guard on feed initiation.

- 242,934 bars on a 30-minute grid, 2005-01-02 to 2024-04-30.
- Training window $W = 24,000$ bars; the walk begins at bar 24,000.
- 218,909 bars scored after dropping non-positive truth and non-finite features. The scored row set is *identical* across arms, so every comparison is bar-for-bar.
- Prediction is per-bar OLS by rank-2 inverse updates, Duan-smeared: $\widehat{RV}_t = ((A_t\hat{\beta})^2 + \hat{\sigma}^2)b_t$.
- $QLIKE = \rho - \log \rho - 1$, $\rho = RV_t / \widehat{RV}_t$. Inference is Diebold–Mariano with Newey–West HAC and the Harvey–Leybourne–Newbold correction. Eras are eight equal blocks.

Every arm below keeps the same twelve lags and the same nine calendar columns. Arms differ *only* in the projection applied to the twelve rungs.

2 What a weight over rungs means on the raw lags

This is the conceptual core and it is elementary, but it was not stated in the earlier note and it reframes every result that follows.

Rung j is the block mean $M_j = L_j^{-1} \sum_{i=1}^{L_j} y_{t-i}^2$ with $L_j = 2^j$. A weighted sum of rungs is therefore a quadrature rule on the lag axis with piecewise-constant weights, and in the variance domain the map is exact:

$$\sum_j w_j M_j = \sum_i \kappa(i) y_{t-i}^2, \quad \kappa(i) = \sum_{j: 2^j \geq i} \frac{w_j}{2^j}.$$

Each raw lag i is collected by every rung long enough to contain it, discounted by that rung's own $1/L_j$. Setting $w_j \propto L_j^{-\alpha}$ makes the inner sum geometric with ratio $2^{-(\alpha+1)}$ from $j_0 = \lceil \log_2 i \rceil$, giving

$$\kappa(i) = \frac{2^{-j_0(\alpha+1)}}{1 - 2^{-(\alpha+1)}} \propto i^{-(\alpha+1)}.$$

A power law of exponent α over rungs is a power law of exponent $\alpha + 1$ over raw lags; the extra power is the $1/L_j$ normalisation of the block means. Since an ARFIMA(0, d , 0) has AR(∞) weights decaying as $i^{-(d+1)}$, the rung exponent *is* the long-memory parameter.

The identity is asymptotic. With twelve rungs truncated at 2048 the measured exponent runs above $\alpha + 1$, because truncation removes long-lag weight and steepens the effective decay—most where small α puts the most mass:

α	predicted $\alpha + 1$	measured	half-weight rung
0.00	1.00	1.189	32
0.25	1.25	1.389	8
0.50	1.50	1.597	2
1.00	2.00	2.032	1
2.00	3.00	2.946	1

So the *measured* exponent is the quantity to read, not the nominal α . The difference matters: it moves the implied d at the optimum from 0.25 to ≈ 0.4 .

3 Result: one column matches twelve rungs

3.1 Three independent routes

route	how the single column is obtained	vs r=12	t
placebo mean	average over nine estimation windows	+0.00001	—
pooled	dominant direction of eight windows	−0.00054	−1.33
<i>a priori</i>	fixed shape, no estimation at all	+0.00034	+1.65

All three are indistinguishable from the twelve-rung ladder. The eight-window placebo mean has SE 0.00052, so it is 0.02 SE from zero.

3.2 Why: the ladder is not twelve-dimensional

The rungs are nested moving averages of a single series, so they are close to duplicates. Averaged over the nine estimation windows, the ladder covariance spectrum is

0.728, 0.128, 0.057, 0.030, 0.018, 0.011, 0.008, 0.006, 0.005, 0.004, 0.003, 0.002

in variance shares—an entropy effective rank of **2.81** of 12, with 72.8% in the top direction. Discarding eleven columns discards little.

Some of the tie is also a genuine variance saving: $\text{tr}((A'A)^{-1})$ is 9.33×10^{-2} at rank 12 against 6.54×10^{-4} at rank 1, a factor of 143 from collinearity. Mean in-sample prediction variance still scales only as p/n (22 against 11 over 24,000 rows), so the bias–variance trade here is real but bounded, not the main effect.

3.3 Which column, and the one parameter that decides

The pooled direction is a smooth near-linear taper in $\log_2$ rung, not a power law:

rung	1	2	4	8	16	32	64	128	256	512	1024	2048
pooled	.437	.425	.396	.366	.329	.289	.255	.212	.162	.096	.030	.005
$k=0$	.474	.421	.361	.344	.091	.204	.282	.037	.304	−.054	−.272	.221

Half the pooled weight lies below rung 8. It is 11.3° from $r^{-0.25}$ and 44.0° from r^{-1} . The published window’s direction, on the second row, is jagged with sign flips—the amplified-tail artifact of §4.4, visible directly in the loadings.

Sweeping fixed shapes, each scored on all 218,909 bars with no estimation whatsoever:

shape	half	κ exp	QLIKE	vs $r=12$	eras
lintaper ^{1.5}	8	1.913	0.13406	−0.00032	6/8
lintaper $\times r^{-0.1}$	4	1.750	0.13409	−0.00029	5/8
lintaper ²	4	2.165	0.13428	−0.00010	4/8
power a=0.30	4	1.430	0.13437	−0.00001	5/8
power a=0.25	8	1.389	0.13470	+0.00032	4/8
power a=0.40	4	1.513	0.13474	+0.00037	4/8
lintaper	8	1.666	0.13496	+0.00058	3/8
power a=0.20	8	1.348	0.13551	+0.00113	2/8
power a=0.15	16	1.308	0.13692	+0.00254	0/8
power a=0.10	16	1.268	0.13904	+0.00466	0/8
base-2 ($r=12$)	12	—	0.13438	—	—

The best arm is an argmin over eleven shapes and is not reported as a win. Bonferroni takes its p from 0.032 to 0.35; the runner-up goes to 0.63. §6 settles it on a split instead.

What *is* robust is the family ordering, because it is not selected: kernels with measured exponent ≤ 1.35 ($d \leq 0.35$) lose by +0.00113 to +0.00466 in 0–2 of 8 eras; everything from 1.39 upward ties. Too little memory is decisively wrong. Reading measured exponents, the best power arms sit at $\kappa \approx 1.39$ –1.43, i.e. $d \approx 0.39$ –0.43, agreeing with the standard long-memory estimate for realised variance.

Note where $\alpha = 0$ sits. Equal weights give $\kappa \propto i^{-1}$, the knife-edge HAR implicitly targets—Corsi’s argument is that a few block averages *mimic* hyperbolic decay—and it is the worst arm tested (+0.01148). This data wants decay strictly faster than hyperbolic.

4 Retraction: the rank-3 win was one window

The earlier note reported three supervised coefficients beating twelve rungs by 0.00080 ($t = -3.30$), with the projection estimated once on rows $[0, W)$ and applied through 2024. Four tests, run in this order, close it.

4.1 Refit cadence does not rescue it, and there is no drift to track

Rebuilding the basis causally from the trailing window loses at every cadence (+0.00023 to +0.00027). Sweeping an exponential halflife on the moments the basis is built from traces the whole bias–variance curve between a fully local refit and an expanding window that never forgets:

arm	QLIKE	vs $r=12$	t
sup frozen(first W)	0.13358	−0.00080	−5.48
base-2 ($r=12$)	0.13438	—	—
sup $h=6000$	0.13453	+0.00015	+2.70
sup $h=24000$	0.13454	+0.00016	+2.72
sup $h=96000$	0.13455	+0.00017	+2.54
sup $h=\infty$	0.13456	+0.00018	+2.60

The curve is **flat**: 0.00003 of spread across a $32\times$ range of memory, no interior optimum, so there is no drift timescale to find. That splits the published number exactly—rank-3 *as a dimension* costs +0.00016, and the specific window is worth −0.00096 relative to any principled rank-3 estimate. None of the effect is “three directions”.

The $h = \infty$ arm is the control the original design lacked. Its information set strictly *contains* the freeze’s at every boundary past the first, and with no drift more data cannot give a worse estimate. It forecasts 0.00098 worse.

4.2 The nine-window placebo

The direct test is to repeat the published procedure on windows nobody chose: freeze on rows $[kW, (k+1)W)$ for every k the sample supports and score all arms causally on the same late region, with identical trailing estimation, so only the projection differs.

k	window	QLIKE	vs $r=12$	t
0	$[0, 24k)$ published	0.13291	−0.00194	−9.72
1	$[24k, 48k)$	0.13537	+0.00052	+4.11
2	$[48k, 72k)$	0.13495	+0.00010	+1.21
3	$[72k, 96k)$	0.13502	+0.00017	+1.93
4	$[96k, 120k)$	0.13519	+0.00034	+3.01
5	$[120k, 144k)$	0.13492	+0.00007	+1.01
6	$[144k, 168k)$	0.13520	+0.00035	+2.36
7	$[168k, 192k)$	0.13490	+0.00006	+0.72
8	$[192k, 216k)$	0.13496	+0.00011	+1.55
—	base-2 ($r=12$)	0.13485	—	—

The eight placebos cluster in $[0.13490, 0.13537]$: mean +0.00022, sd 0.00016, **none** beating the ladder. The published window sits 0.002 outside the pack—13.5 placebo standard deviations. Best-of-nine by chance does not look like that, so the win is *not* a lucky draw. It is also not reproducible by any principled estimator, which is the operative point.

4.3 It is not curvature either

A competing reading is that the twelve rungs lie near a curved three-dimensional surface, which would make the refit instability a rotating tangent rather than noise. Adding polynomial terms in the three scores tests this directly, and they hurt monotonically in degree:

There is no manifold. The structure that exists is flat, and §3 says it is one-dimensional.

arm	cols	QLIKE	vs r=12
linear $r=3$	3	0.13345	−0.00091
base-2 (r=12)	12	0.13436	—
quadratic $r=3$	9	0.13413	−0.00024
cubic $r=3$	19	0.13478	+0.00041
quadratic on raw rungs	90	0.13662	+0.00226

A permutation test on subspace rotation agrees. Conditioning on *zero* training overlap—consecutive windows share 75% of their rows, so lags 1–3 are mechanical—and testing the lag profile against a window-order permutation null, the published estimator shows no significant drift ($p = 0.84, 0.082, 0.067$ per direction; $p = 0.46, 0.18, 0.16$ in the fully disjoint design).

4.4 Nothing imposed recovers it

Why is window $k=0$ different? It is the flattest long-horizon stretch in the sample: smallest minimum eigenvalue of any window (7.78×10^{-4} against $1.15\text{--}4.23 \times 10^{-3}$) and flattest long rungs ($\text{sd}(2048)/\text{sd}(1) = 0.124$ against $0.17\text{--}0.36$). So C^{-1} amplifies the low-variance tail hardest there, and only that window manufactures a long-memory contrast. Note $\text{cond}(C)$ is *not* the discriminator (412 against 331–408).

If that contrast were a real feature, imposing one would find it. It does not:

arm	vs r=12	t
$k=0$ (published), 3 cols	−0.00080	−5.48
$k=0$ direction 1 only, 1 col	−0.00047	−3.15
placebo consensus, 3 cols	+0.00041	+4.53
imposed $r^{-1} + \lg^2$	+0.00086	+12.30
$k_0 d_1 +$ consensus d_2, d_3	+0.00087	+5.99
imposed log-polynomial	+0.00096	+12.21
consensus $d_1 + k_0 d_2, d_3$	+0.00108	+4.37
reverse-engineered from $k=0$ ’s own loadings	+0.00108	+9.52
imposed power	+0.00119	+12.35

Every *a priori* shape loses, including one hand-copied from the winning window’s loadings. Both direction-swaps fail, so the win is not separable into a direction-1 part and a directions-2,3 part—it exists only in that exact estimated triple. (The swaps are re-orthonormalised, so they establish non-separability, not its cause.)

One arm survives and it is the subject of §3: $k=0$ ’s first direction alone, one column, −0.00047 in 6 of 8 eras. At *rank 1* the placebo distribution straddles zero (mean +0.00001, sd 0.00147, 5 of 8 beating the ladder) and $k=0$ is only 1.1 sd from its mean—best of nine but an ordinary draw. **The rank-1 edge of that window is chance; the rank-3 triple is not.** The two ranks give opposite verdicts, and that is the resolution.

5 Corrections to the earlier note

1. **“The compression claim is withdrawn”** was too broad. The *rank-3 superiority* claim is withdrawn. The compression claim is not: one column matches twelve rungs, by three routes (§3).
2. **“Direction 1 is stable to 1–2°”** was wrong as stated. That figure is the smallest principal angle between three-dimensional *subspaces*. The direction-1 *vectors* differ across windows

by 30.1° on average (13.9 to 42.8). The subspaces share a direction; the first Krylov vectors do not.

3. **Coefficient variance is not negligible at rank 12.** $\text{tr}((A'A)^{-1})$ is $143\times$ larger there than at rank 1. The trade is bounded, not absent.
4. **The r^{-1} imposed decay endorsed elsewhere in this project is the second-worst rank-1 arm** (+0.00989). The shape that ties the ladder decays far more slowly.
5. **“It is a property of the estimator, not a three-dimensional structure in the data”** was right in conclusion and wrong in evidence. The seed ladder alone cannot distinguish an estimator artifact from an unreproducible feature; the placebo can, and it shows the published window is an outlier rather than a draw.

Two conceded caveats from the earlier note still stand: the frozen arm moves by 0.00021 across implementations differing only in NaN handling, and this is not a reproduction of the battery (different training clip and panel).

6 The pre-registered test

The -0.00032 of §3 is an argmin over eleven shapes and cannot be reported. The fix is a split, not a correction factor: twenty-one candidate shapes are scored on rows $[24,000, 130,000)$, the best is taken, and that *one* shape is then tested on rows $[130,000, n)$. The regions are disjoint, so no row that chose the shape judges it, and one arm tested once has no multiplicity to correct. Widening the grid from eleven to twenty-one costs nothing here, because the reported statistic never sees the grid.

6.1 Selection: every shape loses

On the selection region the twelve-rung ladder scores 0.12401 and **all twenty-one shapes are worse**, the best by +0.00054:

shape	half	κ exp	QLIKE	vs r=12
lintaper ^{1.5} (selected)	8	1.913	0.12455	+0.00054
lintaper $\times r^{-0.10}$	4	1.750	0.12461	+0.00060
lintaper ^{1.0}	8	1.666	0.12475	+0.00074
power a=0.25	8	1.389	0.12480	+0.00079
power a=0.60 (worst)	2	1.682	0.12848	+0.00447
base-2 (r=12)	12	—	0.12401	—

6.2 Test: the selected shape wins

arm	QLIKE on rows $[130,000, n)$, 112,909 bars
lintaper ^{1.5} , 1 column	0.14298
base-2 (r=12), 12 columns	0.14409
difference	−0.00111

Eras better: 6 of 8. Leave-one-era-out is **stable**—the differential runs -0.00085 to -0.00136 with t from -3.22 to -4.90 , and no single era carries it. That is the sharpest contrast with the withdrawn rank-3 claim, whose differential collapsed to -0.000027 when one era was deleted.

HAC bw	t	p	bootstrap block	95% CI
0	-5.07	3.9×10^{-7}	100	$[-0.00182, -0.00033]$
10	-4.64	3.4×10^{-6}	1000	$[-0.00209, +0.00012]$
100	-3.35	0.0008	5000	$[-0.00225, +0.00000]$
1000	-1.94	0.052		
5000	-1.92	0.054		

6.3 What this does and does not establish

It passes. One fixed, estimation-free column beats twelve rungs on rows never used to choose it, by -0.00111 , robust to era deletion and to bandwidth out to 100. At the widest bandwidths the statistic is marginal ($p \approx 0.05$) and two of three bootstrap intervals touch zero, so this is a solid result rather than an overwhelming one.

But the sign is era-conditional, and that is the more useful finding. The selection half favours the *ladder* by $+0.00054$; the test half favours the *column* by -0.00111 . Those reconcile exactly with the pooled number of §3:

$$\frac{106,000 \times (+0.00054) + 112,909 \times (-0.00111)}{218,909} = -0.00031 \quad \text{against } -0.00032 \text{ measured.}$$

So the full-sample -0.00032 is not a small uniform edge. It is a $+0.00054$ in the first half netted against a -0.00111 in the second, and reporting only the pooled figure would have hidden a sign change.

Had the split run the other way, the pre-registered test would have failed. We state this rather than leave it implicit: a single split is itself an arbitrary choice, which is cause five of §5 in a new costume. What rescues the exercise is that selection did not *need* the sign—no shape won on the selection region, so the winner cannot have been chosen for winning. Only the *ordering* of shapes transferred across the split, and it transferred cleanly.

The claim we are willing to defend is therefore conditional and precise: in the second half of this sample, one fixed column with induced lag kernel decaying near $i^{-1.9}$ beats a twelve-rung ladder; across the full sample the two are a tie; and in the first half the ladder wins. The compression is real throughout—one column is never far from twelve—but which side of the tie it falls on depends on the era.

Artifacts

analysis/ladder_headline.py	the original rank sweep
analysis/ladder_refit.py	refit cadences, seed ladder
analysis/basis_tracking.py	half-life sweep, $h = \infty$ control
analysis/manifold_drift_test.py	window-order permutation null
analysis/ladder_curvature.py	polynomial terms in the scores
analysis/freeze_placebo.py	the nine-window placebo
analysis/imposed_longmem.py	imposed shapes and direction swaps
analysis/rank1_placebo.py	the placebo re-run at rank 1
analysis/rank1_anatomy.py	loadings, spectrum, variance decomposition
analysis/imposed_taper.py	taper sweep and the rung-to-lag check
analysis/preregistered_taper.py	the split test